

A new dust detection method for photovoltaic panel surface based on Pytorch and its economic benefit analysis

Yichuan Shao^a, Can Zhang^b, Lei Xing^c, Haijing Sun^{a,*}, Qian Zhao^d, Le Zhang^a

^a School of Intelligent Science & Engineering, Shenyang University, Shenyang 110044, China

^b School of Information Engineering, Shenyang University, Shenyang 110044, China

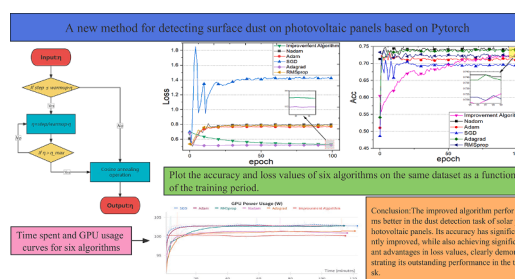
^c School of Chemistry and Chemical Engineering, University of Surrey, GU2 7XH, United Kingdom

^d School of Science, Shenyang University of Technology, Shenyang 110044, China

HIGHLIGHTS

- Optimized and improved the performance of traditional Adam algorithm.
- The addition of Warmup strategy enables stable changes in learning rate.
- Added cosine annealing strategy to make learning rate periodic and avoid overfitting.

GRAPHICAL ABSTRACT



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ABSTRACT

Dust accumulation on the surface of solar photovoltaic panels diminishes their power generation efficiency, leading to reduced energy generation. Regular monitoring and cleaning of solar photovoltaic panels is essential. Thus, developing optimal procedures for their upkeep is crucial for improving component efficiency, reducing maintenance costs, and conserving resources. This study introduces an improved Adam optimization algorithm designed specifically for detecting dust on the surface of solar photovoltaic panels. Although the traditional Adam algorithm is the preferred choice for optimizing neural network models, it occasionally encounters problems such as local optima, overfitting, and not convergence due to inconsistent learning rates during the optimization process. To mitigate these issues, the improved algorithm incorporates Warmup technology and cosine annealing strategies with traditional Adam algorithm, that allows for a gradual increase in the learning rate, ensuring stability in the preliminary phases of training. Concurrently, the improved algorithm employs a cosine annealing strategy to dynamically tweak the learning rate. This not only counters the local optimization issues to some degree but also bolsters the generalization ability of the model. When applied on the dust detection on the surface of solar photovoltaic panels, this improved algorithm exhibited superior convergence and training accuracy on the surface dust detection dataset of solar photovoltaic panels in comparison to the standard Adam method. Remarkably, it displayed noteworthy improvements within three distinct neural network frameworks: ResNet-18, VGG-16, and MobileNetV2, thereby attesting to the effectiveness of the novel algorithm. These findings hold significant promise and potential applications in the field of surface dust detection

* Corresponding author.

E-mail address: seamirror@126.com (H. Sun).

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of solar photovoltaic panels. These research results will create economic benefits for enterprises and individuals, and are an important strategic development direction for the country.

Introduction

As the social economy develops rapidly, the demand for energy consistently rises. Yet, due to the considerable depletion of non-renewable energy sources like oil and natural gas, there's a growing focus on renewable energy sources [1,2]. Solar energy is an inexhaustible renewable energy source for humans, with advantages such as pollution-free, safety, long lifespan, maintenance free, sufficient resources, and potential economic benefits. It plays an important role in long-term energy strategies [3]. The efficiency of solar photovoltaic power generation systems is influenced by many factors such as the material type, layout spacing, area, orientation, environment, and surface dust of solar photovoltaic panels. Surface dust is the most common factor affecting the performance of solar photovoltaic panels [4–6]. The study by Santhakumari et al. [7] shows that due to the size of dust particles being larger than the wavelength of radiation emitted by the sun, the incident solar radiation decreases the amount of incoming light into the battery panel due to dust scattering. At the same time, the dust on the surface of the battery board can cause light reflection, thereby reducing the output efficiency of the battery board. The accumulation of dust on the surface of photovoltaic panels can cause changes in the electrical characteristics of the panel array, leading to reverse bias of the photovoltaic panels and further leading to power loss [8]. This loss will dissipate in the form of heat, leading to uneven heating of photovoltaic panels and posing safety hazards. Therefore, it is very important to use accurate and efficient methods to detect the dust accumulation on the surface of solar photovoltaic panels, which is conducive to timely cleaning of the panels and ensuring their safe and efficient operation [9].

However, at the macroeconomic level, by 2020, the global installed capacity of solar photovoltaics has reached 707.5 gigawatts, with China accounting for 35.8 % [10]. If the cleanliness of the surface of solar photovoltaic panels cannot be effectively maintained, it will result in huge economic losses. The adhesion of dust on the surface of solar photovoltaic panels may have a series of impacts on the economy: the decline in the performance of photovoltaic panels will directly affect the energy generation efficiency of the solar system, thereby affecting the entire energy supply chain; The performance degradation caused by dust adhesion can lead to an increase in production costs per unit of energy, which to some extent affects the competitiveness of solar energy and may even affect the development of the entire renewable energy industry. Due to the fact that the photovoltaic power generation industry is a huge industry, a large amount of cost will be invested to clean the surface of photovoltaic panels. Although it may create some employment opportunities, if the dust detection efficiency is not high, it is more likely to increase operating costs.

Overall, the impact of dust adhesion on the surface of solar photovoltaic panels on the macroeconomy mainly depends on the proportion of solar energy in the national or regional energy mix and the severity of dust problems. In some areas that rely on solar energy, this may become an important economic issue that requires comprehensive consideration of multiple factors such as technology, policies, and markets. To mitigate these impacts, operators of solar systems may need to take measures such as regularly cleaning the surface of the panels, using dust coatings, or using self-cleaning techniques. Although these measures may improve the impact of dust adhesion, the most effective and direct way is to improve the operational efficiency and economy of solar systems by improving the efficiency of dust detection on the surface of photovoltaic panels, promoting the development of the renewable energy industry, and generating positive economic impacts in a sustainable direction.

At present, the main methods for detecting surface dust on solar photovoltaic panels include object detection, image segmentation and

instance segmentation, super-resolution image generation, multispectral and thermal infrared imaging, and deep learning methods. Among them, algorithms such as YOLO [11,12], Faster R-CNN [13], and RetinaNet [14,15] in object detection methods can accurately mark the position and boundary of solar photovoltaic panels in the image, but due to the need for a large amount of computing resources, they have high requirements for hardware and environment. The U-Net [16] and Mask R-CNN [17] algorithms in image segmentation and instance segmentation methods can more accurately identify the position and status of solar photovoltaic panels in the detection and maintenance of solar photovoltaic panels. However, these algorithms require a large amount of computational resources, high real-time requirements, and may also generate a large number of parameters and complex results, increasing the difficulty of maintenance. In recent years, deep learning has achieved significant results in fields such as computer vision and natural language processing. Some scholars have begun to study the application of deep learning algorithms in dust detection of solar photovoltaic panels and have achieved good results, improving the accuracy of detection and classification, such as ResNet [18], EfficientNet [19], and MobileNet [20]. However, this improved deep learning method for neural network frameworks generally has a large number of parameters and training data, which can lead to lower training efficiency [21,22]. The Adam algorithm is a gradient descent algorithm used to optimize neural network training [23]. It combines the advantages of gradient descent and momentum methods, with the aim of adaptively adjusting the learning rate. However, the Adam algorithm occasionally faces issues such as falling into local optima and high computational complexity. The Lion algorithm proposed by Chen et al. [24] uses symbolic functions to handle update volume, which to some extent reduces computational complexity. However, it does not perform as well as Adam in small batch situations; In response to the problem of slow convergence of the Adam algorithm, Liu et al. [25] proposed an improved Adam algorithm that combines adaptive coefficients based on random block coordinate descent and composite gradients; Xie et al. [26] used a new Nesterov momentum estimation (NME) method to estimate the first and second moments in adaptive gradient algorithms to accelerate convergence. However, these improvements might still encounter challenges like inappropriate adjustment of learning rates, increased computational complexity, and local optima.

This paper proposes a combined optimization method that integrates both Warmup and cosine annealing strategies into the Adam algorithm for detecting surface dust on solar photovoltaic panels. Firstly, using the Warmup technology, we incrementally raise the learning rate, allowing the model to more effectively navigate the parameter space during the early stages of training. Subsequently, the cosine annealing strategy is employed to dynamically modify the learning rate, preventing an untimely reduction and ensuring more stable model convergence in the later stages of training. Comprehensive experimental verifications of the improved Adam algorithm were conducted on the surface dust detection dataset for solar photovoltaic panels. The results reveal that the novel algorithm offers notable improvements in convergence speed and generalization capability compared to both the traditional Adam algorithm and other refined versions. The improved Adam algorithm, in which Warmup and cosine annealing are integrated, effectively addressed limitations of conventional Adam algorithms. Demonstrative experiments on the surface dust detection dataset of solar photovoltaic panels highlighted the enhanced performance of the novel algorithm and its potential application in energy-related domains.

Improved algorithm design

Adam optimization algorithm

Adam is an adaptive learning rate optimization algorithm based on gradient descent, with the main idea of adjusting the learning rate by calculating the first-order moment estimation and second-order moment estimation of the gradient. The first order moment estimation is the average value of the gradient, while the second order moment estimation is the average value of the square of the gradient. The Adam algorithm, as one of the commonly used optimization algorithms in neural network models, has the advantage of being able to adaptively adjust the learning rate and adopt different learning rate sizes for different parameters, thus better adapting to the characteristics of different parameters. The Adam algorithm can also adaptively adjust the size of the learning rate, avoiding the hassle of manually adjusting the learning rate. The Adam algorithm can also handle sparse gradients, online settings, and non-stationary settings very well [23]. The Adam algorithm is shown in Algorithm 1:

Warmup

Warmup was first proposed by He et al. [18] in 2016 as a solution for very deep neural networks to prevent overfitting due to high learning rates. The basic idea of Warmup is to use a small learning rate in the early stages of training, gradually increasing to the set initial learning rate within a certain number of iterations. This helps the model to better explore the parameter space in the early stages of training, avoiding oscillation or divergence problems caused by excessive learning rate. The specific Warmup strategy can be set based on specific problems and models, and common methods include linear Warmup and exponential Warmup. Linear Warmup linearly increases the learning rate within a set number of iterations, while exponential Warmup gradually adjusts the learning rate through exponential growth.

The main purpose of using Warmup is to improve the stability and convergence speed of the model in the early stages of training. In the initial stage, the parameters of the model may be in a random initialization state or in a poor region, and using a higher learning rate may cause the model to oscillate or fail to converge. This is because an excessively high learning rate may lead to excessive parameter updates, making the model unable to steadily move towards the optimal solution. To address this issue, Warmup technology was introduced, which allows for the use of a smaller learning rate in the early stages of training and gradually increases the learning rate to the set initial value. The advantage of this method is that it allows the model to undergo a stable exploration in the early stages of training, thereby avoiding instability caused by excessive learning rate. As the training progresses, the learning rate gradually increases, and the model gradually enters a more suitable state. Then, the set maximum learning rate can continue to be used for training. This strategy helps to balance the stability and training speed of the model, improving the efficiency of training and the possibility of successfully converging to the optimal solution.

The initial Warmup strategy formula is shown in formula (1):

$$\eta = \eta_{\text{base}} \frac{\text{step}_{\text{current}}}{\text{epoch}_{\text{Warm}} \text{step}_{\text{One_epoch}}} \quad (1)$$

In formula (1), η_{base} represents the initial learning rate, the numerator $\text{step}_{\text{current}}$ in the fraction represents the number of steps taken in the current stage of the training process, the denominator $\text{epoch}_{\text{Warm}}$ represents the number of epochs set for the Warmup process before the training begins, and the denominator $\text{step}_{\text{One_epoch}}$ represents the number of steps required to train each epoch during the training process, which is determined by the size of the data and the batch size.

We have made some modifications to the original formula, and the modified formula is shown in formula (2):

$$\eta_{\text{Warmup}} = \eta \frac{\text{step}_{\text{current}}}{\text{step}_{\text{Warmup}}} \quad (2)$$

The denominator in the fraction represents the number of preheating steps set during the initialization phase, which can effectively reduce the time required for the entire preheating process.

Cosine annealing strategy

The main idea of the cosine annealing strategy is to simulate the annealing process of the cosine function, periodically changing the learning rate during the training process, so that the model can better make fine adjustments in the later stage of training and improve convergence performance. The original formula for cosine annealing strategy is shown in formula (3):

$$\eta = \eta + \frac{1}{2}(\eta_{\text{max}} - \eta) \left[1 + \cos\left(\frac{T}{T_i} \pi\right) \right] \quad (3)$$

The T in the numerator represents the total number of training cycles to the current stage, and the T_i in the denominator represents the set cosine annealing cycle. Based on the ratio between T and T_i , the learning rate exhibits a cosine like variation.

This article improves the original cosine annealing strategy by calculating the learning rate using the relationship between the current number of steps and the set warmup value. The improved formula is shown in formula (4):

$$\eta \leftarrow \eta + \frac{1}{2}(\eta_{\text{max}} - \eta) [1 + \cos(\text{step} - (\text{warmup}\lambda / T_0)\pi)] \quad (4)$$

The values of warmup and T_0 are given during initialization, while λ represents lr_{max}/lr , which is the ratio of the maximum learning rate to the current learning rate.

The unique feature of cosine annealing strategy is that it causes the learning rate to periodically change between the maximum and minimum learning rates according to the annealing curve of the cosine function. The advantage of this strategy is that it can improve the stability of the model in the later stage of training, prevent oscillations caused by high learning rates, jumping out of local optima, and slow training speed caused by low learning rates.

Compared with traditional fixed learning rates, cosine annealing algorithm can achieve faster convergence because it can use a smaller learning rate to fine tune model parameters in the later stages of training. This strategy helps to prevent the oscillation problem caused by the rapid descent speed of the gradient, thereby improving the training stability and generalization performance of the model.

Another advantage is that the cosine annealing algorithm only needs to set a small number of hyperparameters, such as maximum learning rate, maximum number of iterations, and minimum learning rate, making it easier to adjust and optimize the training process of the model. This makes it a widely used learning rate scheduling strategy in practical applications.

Improvement algorithm

When using optimization algorithms, learning rate is a key parameter that requires consideration of multiple factors. Usually, different learning rate values are set at different training stages to adapt to the progress of training. The technology adopted by the Adam algorithm is to ensure that the learning rate does not decay too quickly while adaptive gradient adjustment. To achieve this goal, the Adam algorithm uses exponential weighted averaging for gradients and the square of gradients, which is actually a cumulative calculation of historical gradients, which helps to reduce the impact of historical gradients on the current gradient.

In the initial stage of training, the convergence speed of the Adam

algorithm is very fast due to the acceleration of the first-order momentum of the gradient. However, this also leads to significant oscillations in the Adam algorithm near the optimal value, exhibiting significant instability. In addition, when the Adam algorithm falls into a local optimal state, it is difficult to jump out of this local optimal state.

These issues indicate that learning rate scheduling strategies are crucial for optimizing the performance of algorithms. Appropriate learning rate scheduling can help balance fast convergence and avoid instability issues. Strategies such as cosine annealing are a method of dynamically adjusting the learning rate to balance these factors at different stages of training, thereby improving the stability and convergence of the algorithm.

During the training process, by comparing the current number of training steps with the pre-set number of Warmup steps, it is determined whether to use the Warmup strategy to gradually increase the learning rate or use the cosine annealing strategy to periodically change the learning rate. This improved approach fully considers the needs of different stages of training.

Through this improvement, the learning rate can be rapidly increased in the early stages of training to accelerate the convergence speed of the model. In the subsequent stages of training, through detailed learning rate adjustments, the model parameter space can be searched more accurately, thereby improving the accuracy of the model. This strategy effectively improves the generalization ability of the model and helps to avoid falling into local optima.

Overall, this improved method combining Warmup and cosine annealing helps to balance the speed and accuracy of model training, making the training process more stable and efficient. The flexibility of this strategy can be adjusted based on specific tasks and data to optimize the performance of the model.

From small to large: In the initial stage, a higher learning rate helps the model learn useful features and parameter adjustments faster. But as the training progresses, the distribution of the model gradually stabilizes. If a higher learning rate is continued at this time, it may cause the model to fluctuate significantly near the optimal solution, making it difficult to converge. Therefore, reducing the learning rate is to search for the optimal solution space more precisely, enabling the model to stably adjust parameters near the optimal solution.

Changing from large to small: In the initial stage, the weights of the model are randomly initialized. If a larger learning rate is chosen at this time, it may lead to unstable situations in the model, such as oscillation or divergence. By using a preheating learning rate, the learning rate can be set to be small in the early stages of training, allowing the model to gradually stabilize. At this small learning rate, the model can gradually learn useful features and parameter adjustments in a stable environment, thereby better adapting to tasks. Once the model reaches a relatively stable state, gradually increasing the learning rate can help the model converge to the optimal solution faster, as the model has already undergone a certain degree of preheating at a lower learning rate. This strategy helps to balance training speed and model stability, and is one of the commonly used techniques in many deep learning tasks. By carefully adjusting the size of the preheating learning rate and the training steps, the training process of the model can be better controlled, thereby improving the efficiency of training and the possibility of successfully converging to the optimal solution.

This strategy of gradually reducing learning rate is very effective in practice, as it fully considers the changes in the model's state during the training process. By using a smaller learning rate in the later stages of training, the model can approach the optimal point more accurately, thereby improving its generalization performance and convergence speed. This is also an important reason why many optimization algorithms and learning rate scheduling strategies adopt this approach to balance training speed and model stability. The improved algorithm is shown in Algorithm 2:

The flowchart for selecting whether to use Warmup or cosine annealing strategy is shown in Fig. 1:

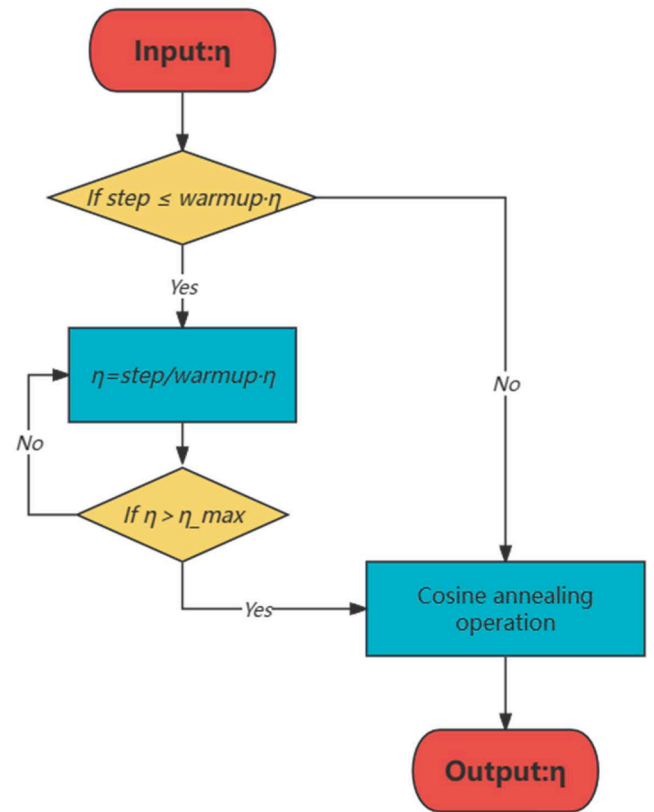


Fig. 1. Warmup and cosine annealing strategy flowchart.

The following is a more detailed explanation of this process:

Initial learning rate(η): At the beginning of the training, input an initial learning rate.

Warmup stage: Conduct Warmup operations in the early stages of training. Firstly, determine whether it is necessary to enter the Warmup stage. If the initial learning rate is less than the set learning rate of the Warmup stage, it enters the Warmup stage. In the Warmup stage, use the given formula to gradually increase the learning rate, so that the model is not affected by excessive learning rate at the beginning of training. This helps to prevent the model from falling into local optima.

Upper limit of learning rate: After Warmup operation, check if the learning rate is less than the set maximum learning rate. If the learning rate is less than the maximum learning rate, continue to process the learning rate in the Warmup stage, otherwise skip out of the Warmup stage.

Cosine annealing stage: If the learning rate jumps out of the Warmup stage, it enters the cosine annealing stage. In the cosine annealing stage, the learning rate is gradually reduced according to the set cosine annealing formula. This helps the model to optimize parameters more stably in the later stages of training, reducing the amplitude of learning rate and preventing jumping out of the optimal solution.

Finally, the learning rate adjustment strategy was processed by Warmup and cosine annealing to generate a reasonable learning rate to balance the fast convergence of the model and the need to avoid falling into local optima during the training process.

Experimental design and result analysis

Selection of improved algorithm learning rate

The improved algorithm aims to enhance the generalization ability of the model and avoid falling into local optima. In order to select the learning rate combination value that optimizes the performance of the

improved algorithm, we conducted comparative experiments on multiple learning rate combinations on the MNIST dataset, and the experimental results are shown in Fig. 2.

In Fig. 2, the legend represents the maximum and minimum learning rates for each combination of learning rates. It can be seen that the combination of parameters with a minimum learning rate of $1e-4$ and a maximum learning rate of $1e-2$ is used to achieve the best effect. The combination of two parameters with a minimum learning rate of $1e-3$ performs worse in the early stages of training than the combination with a minimum learning rate of $1e-4$. This phenomenon may be due to the fact that higher learning rates may lead to model instability in the early stages of training and cause oscillations near extreme points.

Algorithm comparison experiment and analysis

Experimental dataset

In this study, the solar photovoltaic panel dust detection dataset we used was sourced from the widely recognized Kaggle website, and its value lies in its inclusion of two distinct categories. Firstly, we have images of cleaning solar photovoltaic panels, which present a clean state on the surface of the solar panels, free from dust or impurities. In contrast, the second type of image shows the unclean state of the surface of the solar panel, where dust and particulate matter are suspended in the air, so we can clearly see dust flying in the image. In total, this dataset contains 3260 color RGB images, providing rich and diverse training and testing materials for deep learning algorithms to accurately distinguish the cleanliness and pollution status of solar photovoltaic panels. Through this dataset, we were able to conduct experimental research aimed at developing automated image processing algorithms that can effectively detect and classify these two states, providing strong support for quality control and performance improvement in the solar energy industry. Two types of images are shown in Fig. 3:

As shown in Fig. 3, (a) and (b) represent the clean surface of the solar photovoltaic panel, while (c) and (d) represent the dusty surface of the solar photovoltaic panel.

Experimental setup

In order to more accurately evaluate the performance of the proposed improved algorithm, we utilized the latest version of Pytorch 2.0.1 under the Ubuntu operating system to program the algorithm using the PyTorch Lightning framework. In the experiment, we used the NVIDIA GeForce RTX 3080 graphics card to provide excellent computing performance. In order to construct our model, we chose ResNet-18 residual neural network, which is a highly respected model architecture in the field of computer vision.

In terms of data processing, we adopted the solar photovoltaic panel dust detection dataset and divided the data into training, validation, and

testing sets in a strict 7:2:1 ratio to ensure that the quality and quantity of training, validation, and testing data are fully guaranteed. In addition, we have set up 100 epochs to ensure that the model has sufficient training time to achieve optimal performance.

In order to comprehensively evaluate the effectiveness of the improved algorithm, we conducted over 10 comparative experiments using SGD, RMSprop, Adagrad, Nadam, Adam algorithms, as well as our improved algorithm. Finally, we calculated the average of multiple experiments to analyze the differences in performance of these algorithms in the current task. Subsequently, we conducted a detailed comparative analysis of the performance of these six algorithms in key performance indicators such as accuracy and loss value, in order to further verify the superiority of our proposed algorithm.

Experimental results and analysis

The accuracy and loss curves of the six algorithms for this task are shown in Figs. 4 and 5, respectively. Finally, we present the experimental results in the form of specific numerical values in Table 1. The curve graphs of the time required for the six algorithms to complete the entire training and the GPU occupancy rate are shown in Fig. 6.

As shown in Fig. 4, we conducted a trend analysis on the image classification accuracy of the solar photovoltaic panel dust detection dataset. It is obvious that the SGD algorithm performs the most unstable throughout the entire training process, and its accuracy is relatively low. By contrast, our improved algorithm may be slightly inferior to other algorithms in the early stages of training, but as training progresses, its accuracy rapidly increases and shows a significant improvement trend at the end of training. By zooming in locally, we can observe that compared to our improved algorithm, Nadam, Adagrad, and RMSprop show an upward trend, while the other three algorithms show a downward trend.

As shown in Fig. 5, we conducted a detailed analysis of the image classification task and loss value trend of the solar photovoltaic panel dust detection dataset. Like the accuracy curve, the SGD algorithm performs relatively poorly throughout the entire training process, with its loss value remaining at a relatively high level. Contrary to the accuracy curve, our improved algorithm exhibits a lower level of loss value during the initial training period. The loss value curves of Nadam, RMSprop, and Adam algorithms are basically consistent. Although the overall loss value of our improved algorithm is slightly higher than that of the Adagrad algorithm, it shows a decreasing trend throughout the entire training process, even at the end of the training phase, further confirming the effectiveness of the improved algorithm and its excellent performance in dust detection of solar photovoltaic panels.

The final results of the training of the three algorithms are shown in Table 1.

The variation curve of GPU occupancy rate over time for the six algorithms during the entire training process is shown in Fig. 6. The curves

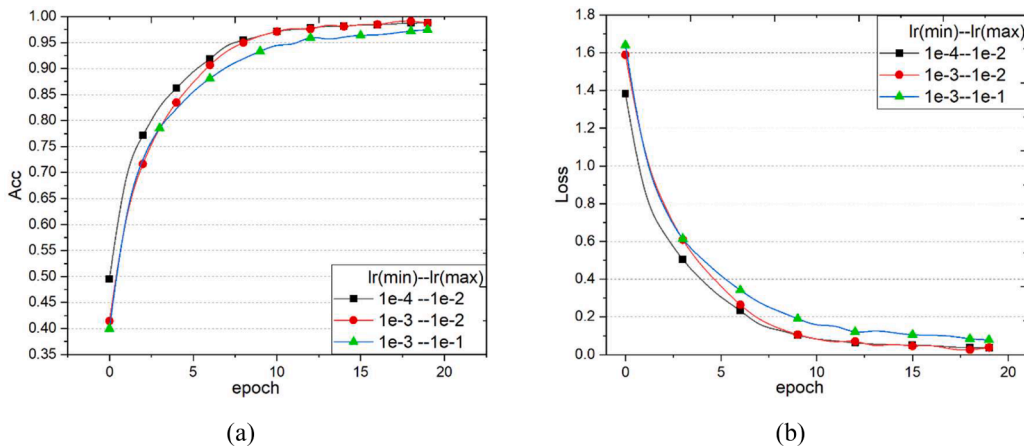


Fig. 2. Comparison of different combinations of learning rate changes on the MNIST dataset.



(a)



(b)



(c)



(d)

Fig. 3. Examples of two types of images.

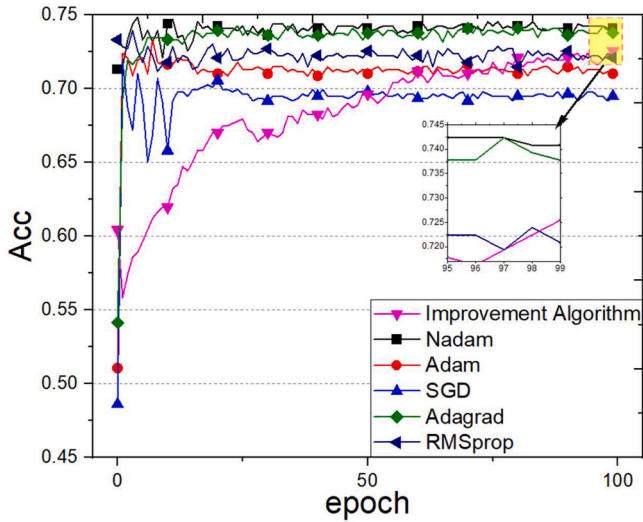


Fig. 4. Detection accuracy of six algorithms.

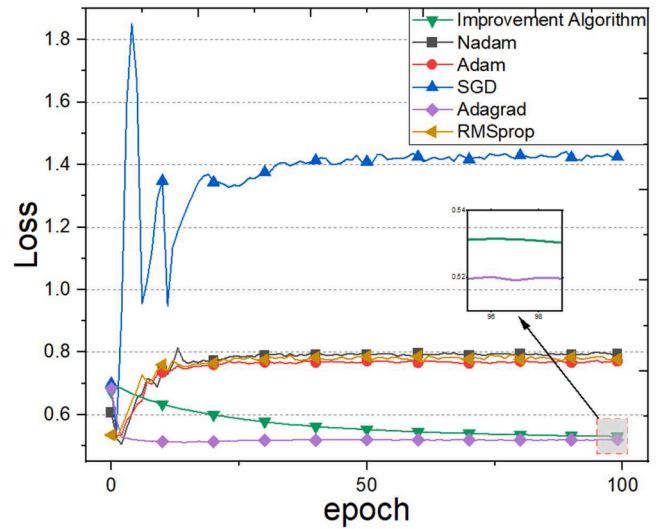


Fig. 5. six algorithms for detecting loss values.

in the figure have been smoothed to observe the differences between the algorithms. Among them, the GPU usage of SGD, RMSprop, and Adam algorithms is relatively high, but the time required for Adam algorithm to complete the entire training is lower than the other two. Our proposed improved algorithm achieved the best performance under GPU usage, indicating that the improved algorithm requires fewer resources to

complete the same training task. Moreover, good results have been achieved in terms of training time.

Based on the analysis of the results, it is evident that there are significant performance differences among these six algorithms in the dust detection task of solar photovoltaic panels. Especially for the SGD algorithm, its performance in this task is the least ideal, with relatively low

Table 1

Training of Three Algorithms for Solar Photovoltaic Panel Dust Detection Datasets.

Algorithm	Accuracy	Loss
Improvement Algorithm	72.45 %	0.5301
Nadam	74.08 %	0.7961
Adam	71.13 %	0.7718
SGD	69.43 %	1.4246
Adagrad	73.77 %	0.5202
RMSprop	72.01 %	0.7773

accuracy and high loss values, indicating that its training effectiveness in this task is limited.

On the contrary, the improved algorithm demonstrated better performance in the dust detection task of solar photovoltaic panels. Its accuracy has significantly improved, while also achieving a significant advantage in loss value, clearly demonstrating its outstanding performance in the task. Moreover, there is still a possibility of improvement in this advantage at the end of the entire training. Especially compared to the standard Adam algorithm, the improved algorithm performs better, further highlighting its significant improvement and outstanding performance in this task.

These results not only provide strong support for the improved algorithm proposed in this study, but also emphasize the significant impact of selecting appropriate optimization algorithms on the performance of deep learning tasks. The successful application of improved algorithms in the dust detection task of solar photovoltaic panels provides useful experience and demonstration for related fields, and provides strong inspiration for further improvement and optimization of

deep learning applications.

Comparative experiment and analysis of network models

In order to compare the performance of improved algorithms in different neural network architectures and highlight the comprehensiveness of the comparative experiment, we conducted experiments on the dust detection dataset of solar photovoltaic panels on three different neural networks: ResNet-18, VGG-16, and MobileNetV2. Due to the improvement of the Adam algorithm in this study, in order to make the comparison of experimental results more intuitive and to save computational resources, we only selected the improved algorithm and the standard Adam algorithm for comparative experiments in this experiment. This series of comparative experiments helps to comprehensively evaluate the performance of the improved algorithm and provides strong support for the reliability of the research results.

Figs. 7, 8, and 9 respectively show the performance of the improved algorithm and the standard Adam algorithm on the loss value and accuracy of the surface dust detection dataset of solar photovoltaic panels in three different neural network models: ResNet-18, VGG-16, and MobileNetV2.

From these charts, we can clearly observe the performance differences under different neural network architectures and the superiority of the improved algorithm over the Adam algorithm. This comprehensive comparative experiment helps to deeply understand the application potential of improved algorithms in different environments and provides important data support for further research.

The experimental results of the improved algorithm and Adam algorithm under three different network models are summarized in

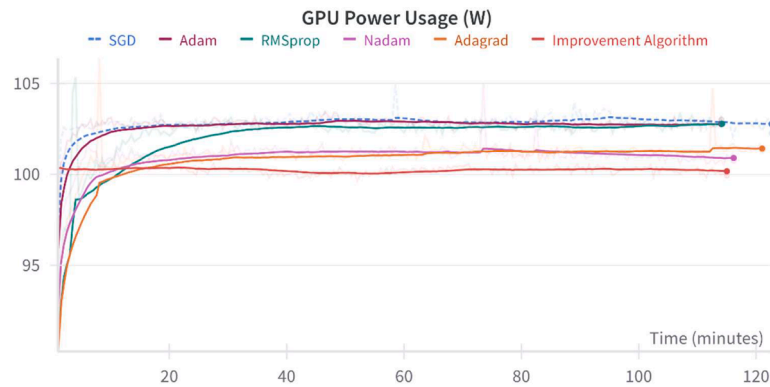


Fig. 6. Time spent and GPU usage curves for six algorithms.

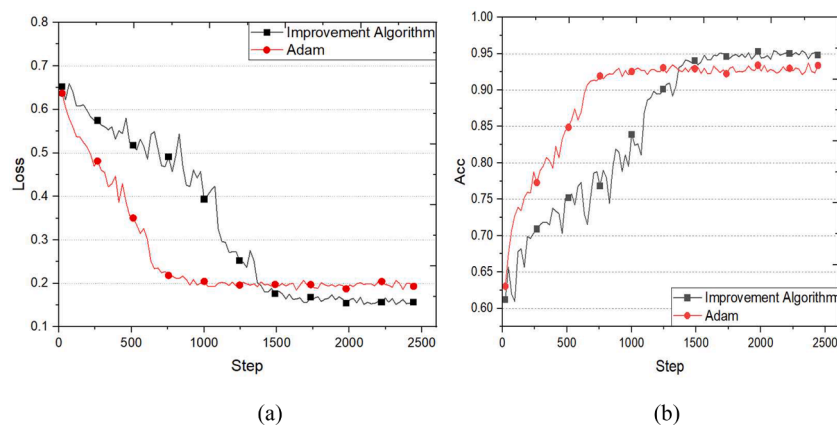


Fig. 7. Performance comparison of different optimization algorithms on the ResNet-18 model. The left figure (a) shows a comparison of loss values, while the right figure (b) shows a comparison of accuracy.

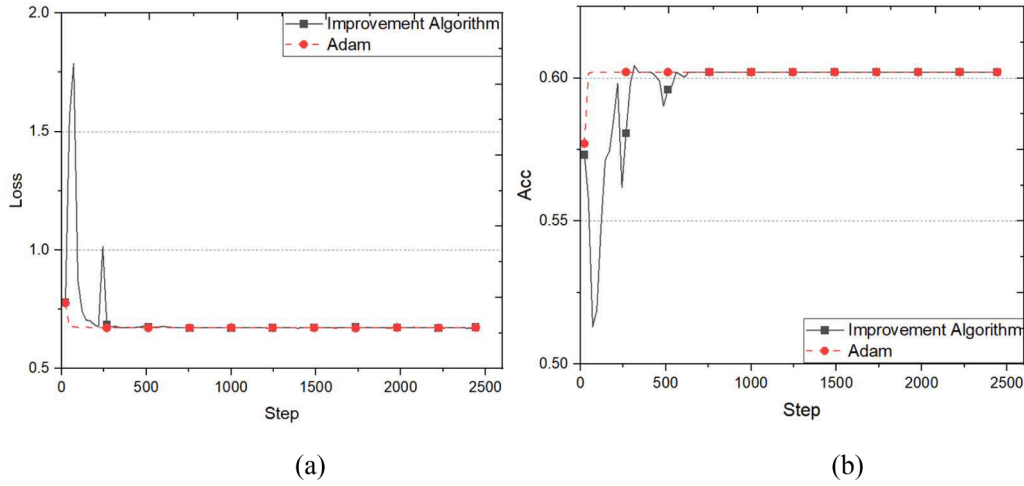


Fig. 8. Performance comparison of different optimization algorithms on the VGG-16 model. The left figure (a) shows a comparison of loss values, while the right figure (b) shows a comparison of accuracy.

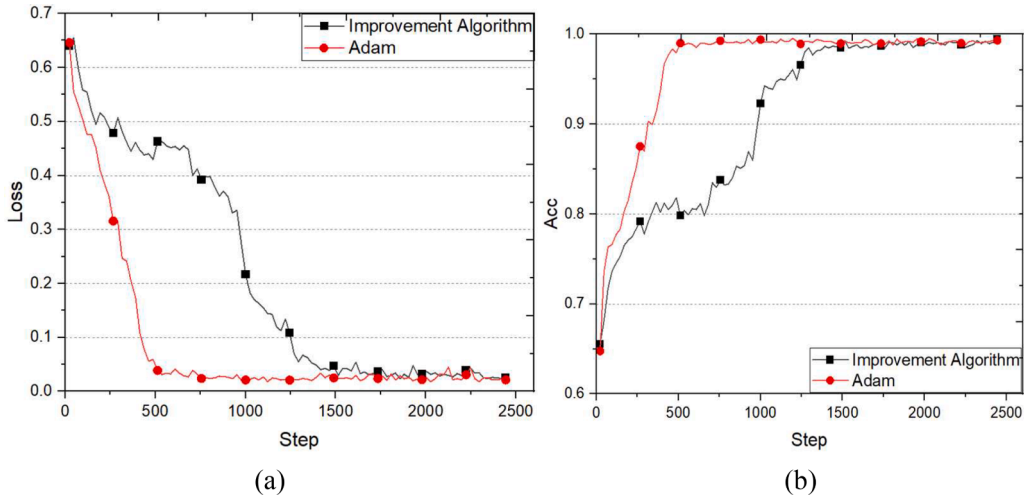


Fig. 9. Performance comparison of different optimization algorithms on the MobileNetV2 model. The left figure (a) shows a comparison of loss values, while the right figure (b) shows a comparison of accuracy.

Table 2.

The experimental results clearly indicate that the performance of the proposed improved algorithm on the surface dust detection dataset of solar photovoltaic panels is far superior to the Adam algorithm and SGD algorithm. Specifically, on the ResNet-18 model, the improved algorithm achieved excellent classification accuracy of 95.12 %, while in contrast, the classification accuracy of Adam and SGD were only 93.38 % and 69.41 %, respectively. On the VGG-16 model, the improved algorithm also showed significant performance improvement, with a classification accuracy of 61.21 %, while Adam's classification accuracy was 60.05 %. On the MobileNetV2 model, the improved algorithm achieved excellent performance, with a classification accuracy of 99.43

Table 2
Training Results of Two Algorithms under Different Network Models.

Algorithm	Network	Accuracy	Loss
Improvement Algorithm	ResNet-18	95.12 %	0.14083
Adam	ResNet-18	93.38 %	0.1932
Improvement Algorithm	VGG-16	61.21 %	0.6741
Adam	VGG-16	60.05 %	0.6811
Improvement Algorithm	MobileNetV2	99.43 %	0.0252
Adam	MobileNetV2	98.08 %	0.0248

Algorithm 1

Adam algorithm.

- 1: **Inputs:** initial point x_0 , step sizes $\{\alpha_t\}_{t=1}^T$, first moment decay β_1 , second moment decay β_2 , regularization constant ϵ
- 2: Initialize $m_0 = 0$ and $v_0 = 0$
- 3: **for** $t = 1$ **to** T **do**
- 4: $g_t = \nabla f_t(x_{t-1})$
- 5: $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$
- 6: $v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$
- 7: $\hat{m}_t = m_t / (1 - \beta_1^t)$
- 8: $\hat{v}_t = v_t / (1 - \beta_2^t)$
- 9: $x_t = x_{t-1} - \alpha_t \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$
- 10: **end for**

%, while Adam's classification accuracy was 98.08 %. This series of results clearly highlights the significant impact of selecting the appropriate optimizer on performance in image classification tasks, and the improved algorithm has demonstrated outstanding advantages in this task, providing solid support for its potential in practical applications.

It can be clearly observed that the improved algorithm performs significantly better than the Adam algorithm on the surface dust detection dataset of solar photovoltaic panels under three different

Algorithm 2

Improved algorithm.

```

1:  Inputs: initial point  $x_0$ , first moment decay  $\beta_1$ , second moment decay  $\beta_2$ ,
    regularization constant  $\epsilon$ 
2:  Initialize  $\mu_0 = 0$  and  $v_0 = 0$ ,  $\eta$ ,  $\eta_{\max}$ , warmup,  $T_0$ 
3:  For  $t = 1$  to  $T$  do
4:     $g_t = \nabla f_t(x_{t-1})$ 
5:     $\mu_t = \beta_1 \mu_{t-1} + (1 - \beta_1) g_t$ 
6:     $v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$ 
7:     $\hat{\mu}_t = \mu_t / (1 - \beta_1^t)$ 
8:     $\hat{v}_t = v_t / (1 - \beta_2^t)$ 
9:     $\lambda \leftarrow \eta_{\max} / \eta$ 
10:   If  $\text{step} \leq \text{warmup} \times \lambda$ 
11:      $\eta \leftarrow \text{step} / \text{warmup} \times \eta$ 
12:   If  $\eta > \eta_{\max}$ 
13:     break
14:   else do
15:      $\eta \leftarrow \eta + \frac{1}{2}(\eta_{\max} - \eta)[1 + \cos(\text{step} - (\text{warmup}\lambda / T_0)\pi)]$ 
16:    $x_t = x_{t-1} - \eta \hat{\mu}_t / (\sqrt{\hat{v}_t} + \epsilon)$ 
end for
Return  $x_t$ 

```

network models, especially in the ResNet-18 network model, where the loss value and accuracy of the improved algorithm significantly exceed the Adam algorithm. It is worth noting that under these three different network models, the improved algorithm may exhibit slower convergence speed in the early stages of training. This may be due to the introduction of Warmup and cosine annealing steps in the improved algorithm, which enables it to jump out of the local optimal solution during the training process, resulting in a slower training speed in the early stage. However, it is precisely due to these improvement measures that the improved algorithm can demonstrate significantly better performance than the Adam algorithm in the later stages of training. These findings emphasize the potential and significant advantages of improved algorithms in image classification tasks, providing strong support for their widespread application in practical applications.

The impact of surface dust detection on improving photovoltaic power generation based on Pytorch

Related studies have shown that the more ash accumulation, the worse the output characteristics of photovoltaic modules [27]. J. M. Ruiz et al. [28] studied the effect of dust obstruction on the power generation efficiency of photovoltaic modules and found that when half of the surface of the photovoltaic module is obstructed, its output power loss is at least 19 %, while when it is completely obstructed, its output power loss can reach up to 79 %. Chen et al. [29] found through experiments at different ash densities that a dust density of $10\text{g}/\text{m}^2$ reduced the output power of photovoltaic modules by 34 %; In addition, the photoelectric conversion efficiency and filling factor (FF) also decrease with the increase of ash density. When the ash density is low (less than $30\text{g}/\text{m}^2$), for every $10\text{g}/\text{m}^2$ increase in dust mass density, the conversion efficiency decreases by an average of 3.4 %. The improved algorithm proposed in this article has significantly improved the efficiency of dust detection on the surface of photovoltaic panels compared to the Adam algorithm, and is suitable for dust detection on the surface of photovoltaic panels in various large photovoltaic power plants.

Conclusion

This study proposes an innovative and improved Adam algorithm variant specifically designed for surface dust detection tasks on solar photovoltaic panels. Compared to the traditional Adam algorithm, this algorithm introduces Warmup and cosine annealing strategies and applies them to the energy field. Specifically, extensive and in-depth validation experiments have been conducted on the surface dust

detection dataset of solar photovoltaic panels. The experimental results clearly demonstrate the effectiveness and excellent performance of the improved algorithm in this field. This algorithm has achieved significant improvement in classification accuracy while successfully reducing loss values, providing solid theoretical and experimental support for its potential in practical applications. The application prospects of this innovative algorithm are particularly significant in the dust detection of solar photovoltaic panels, and it is expected to bring revolutionary impact on quality control and performance improvement in this field.

In addition, the contribution of this study is not limited to the algorithm itself, but also includes deep learning applications in the energy field, providing important support for improving the efficiency and maintainability of solar photovoltaic panels. Future research directions may include verifying the performance of the algorithm in a wider range of applications, exploring the adaptability of different problems and datasets, and further optimizing the algorithm to achieve wider applications. The potential of deep learning in the energy field still needs to be explored, and this study provides useful inspiration and guidance for promoting the application of deep learning technology in this field.

Overall, the improved algorithm proposed in this study has demonstrated excellent performance in surface dust detection tasks of solar photovoltaic panels, demonstrating its potential and practicality in deep learning optimization. This algorithm not only achieves significant advantages under different neural network architectures, but also successfully overcomes the limitations of local optimal solutions and improves generalization performance and robustness by introducing Warmup and cosine annealing strategies. However, there are still many future research directions worth exploring in this study. Firstly, we can consider verifying the performance of the improved algorithm in a wider range of applications to further confirm its universality in practical applications. In addition, adaptability to different dataset sizes and features is also an important research direction. Further research can focus on the automatic adjustment of optimization algorithms to adapt to different problems and datasets, thus achieving wider applicability. In addition, the application potential of deep learning in the energy field is enormous. We encourage future research to continue exploring the application of deep learning technology in the fields of solar energy and other renewable energy to address challenges in energy efficiency, prediction, and monitoring. Such research will make important contributions to the development of sustainable energy and environmental protection.

In summary, this study provides a solid foundation for the application of deep learning algorithms in the dust detection task of solar photovoltaic panels, and also provides useful inspiration for future research. It encourages further exploration and innovation to promote the application of deep learning technology in the fields of energy and environment.

Dear editors

No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed.

Yours sincerely,

CRedit authorship contribution statement

Yichuan Shao: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Funding acquisition. **Can Zhang:** Writing – review & editing, Writing – original draft, Validation. **Lei Xing:** Writing – review & editing, Supervision. **Haijing Sun:** Writing – review & editing, Writing – original draft, Validation, Supervision. **Qian Zhao:** Supervision. **Le Zhang:** Supervision.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

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